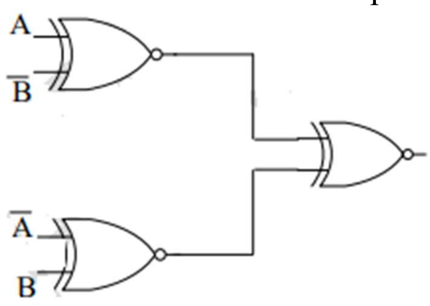
Assignment

1. Convert the following BCD number to decimal
   $010101101001_{bcd.}$

2. Perform the following operations in 2's complement method (use 8 bit representation)
   (a) $(52)_{10} - (23)_{10}$
   (b) $(23)_{10} - (52)_{10}$
   (c) $(52)_{10} - (-23)_{10}$

3. The output of the circuit shown is equal to



(a) 0
(b) 1
(c) $\bar{A}B + A\bar{B}$
(d) $\overline{(A \oplus B)} \oplus (A \oplus B)$

4. The minimized form of the logical expression.
   $(A\,B\,C + A\,B\,C + ABC + AB\,C)$

5. Using 4-variable K'Map simplify the Boolean function given by
   F (w, x, y, z) = Σm (2, 3, 10, 11, 12, 13, 14, 15)

6. Using K'Map simplify in the product-of-sum form the function given by
   F(A,B,C,D} = ΠM (0, 6, 10, 12)

7. The Boolean expression for the truth table shown is

CD \ AB	00	01	11	10
00	1	1	0	1
01	0	0	0	1
11	1	0	0	0
10	1	0	0	1

8. Use Boolean Algebra to show that

$$A'BC'+AB'C'+AB'C+ABC'+ABC = A+BC'$$

9. Derive the logical expression for sum and carry of a full adder.

10. Implement full adder using half adder.